

Stage 1

```
pfSense 2.8.1-RELEASE amd64 20260701-1525
Bootup complete

FreeBSD/amd64 (pfSense.home.arpa) (ttyv0)

VirtualBox Virtual Machine - Netgate Device ID: 979fd4e553fc8158db97

*** Welcome to pfSense 2.8.1-RELEASE (amd64) on pfSense ***

WAN (wan)  -> em0 -> v4/DHCP4: 10.0.2.15/24
              v6/DHCP6: fd17:625c:f037:2:a00:27ff:fe00:d488/64
LAN (lan)   -> em1 -> v4: 192.168.1.1/24
DMZ (opt1)  -> em2 -> v4: 192.168.2.1/24

0) Logout / Disconnect SSH          9) pfTop
1) Assign Interfaces                10) Filter Logs
2) Set interface(s) IP address     11) Restart GUI
3) Reset admin account and password 12) PHP shell + pfSense tools
4) Reset to factory defaults       13) Update from console
5) Reboot system                   14) Enable Secure Shell (sshd)
6) Halt system                     15) Restore recent configuration
7) Ping host                       16) Restart PHP-FPM
8) Shell

Enter an option: █
```

Networks on pfSense:

WAN: 10.0.2.15

LAN: 192.168.1.1

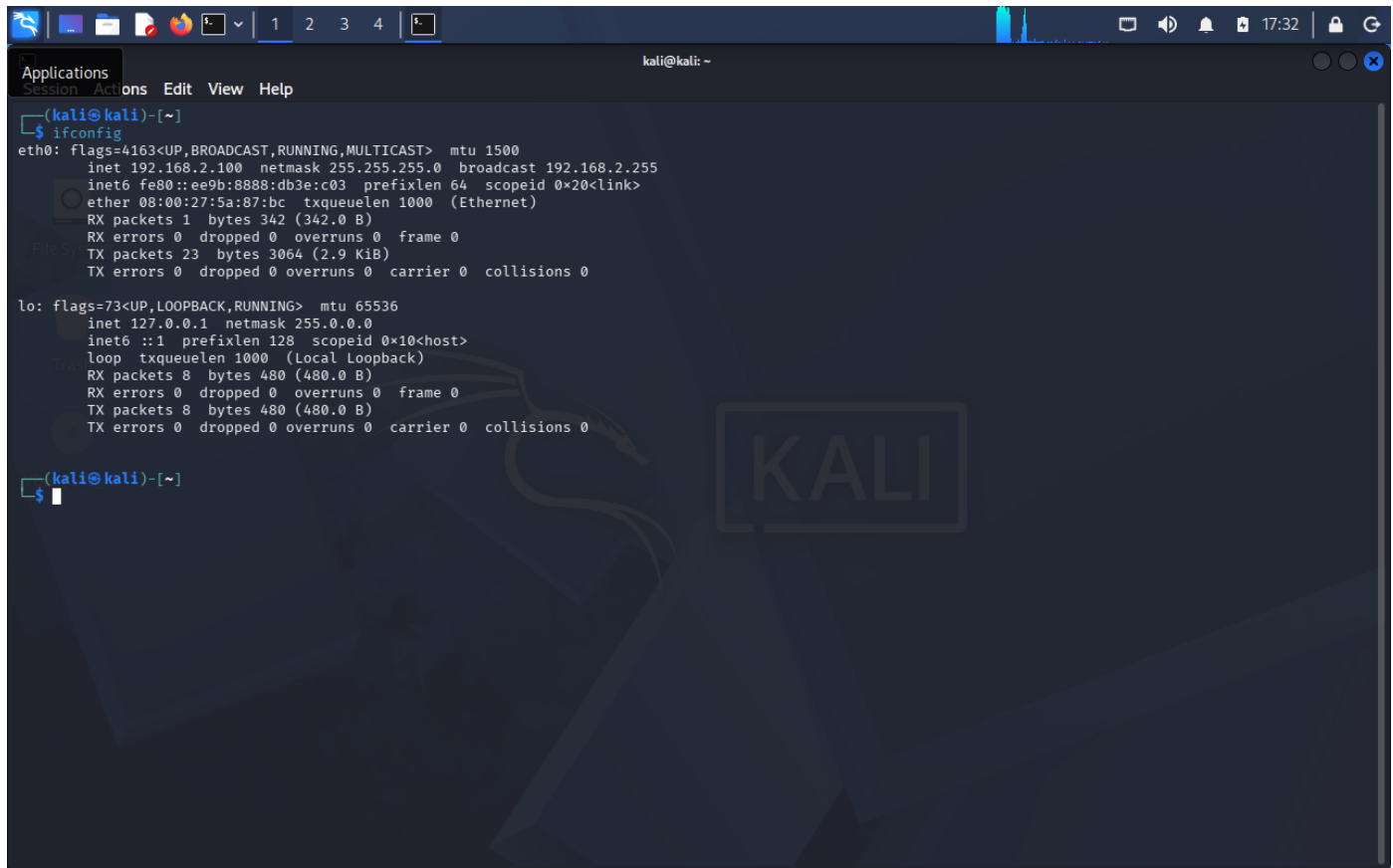
DMZ: 192.168.2.1

```
Aug 15 17:30
ubuntu@ubuntu-VirtualBox: ~
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   link/ether 08:00:27:ce:96:ec brd ff:ff:ff:ff:ff:ff
   inet 192.168.1.101/24 brd 192.168.1.255 scope global dynamic noprefixroute enp0s3
       valid_lft 7046sec preferred_lft 7046sec
   inet6 fe80::a00:27ff:fece:96ec/64 scope link
       valid_lft forever preferred_lft forever
ubuntu@ubuntu-VirtualBox:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
   inet 192.168.1.101 netmask 255.255.255.0 broadcast 192.168.1.255
   inet6 fe80::a00:27ff:fece:96ec prefixlen 64 scopeid 0x20<link>
   ether 08:00:27:ce:96:ec txqueuelen 1000 (Ethernet)
   RX packets 117 bytes 10248 (10.2 KB)
   RX errors 0 dropped 0 overruns 0 frame 0
   TX packets 213 bytes 24864 (24.8 KB)
   TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
   inet 127.0.0.1 netmask 255.0.0.0
   inet6 ::1 prefixlen 128 scopeid 0x10<host>
   loop txqueuelen 1000 (Local Loopback)
   RX packets 519 bytes 42125 (42.1 KB)
   RX errors 0 dropped 0 overruns 0 frame 0
   TX packets 519 bytes 42125 (42.1 KB)
   TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

ubuntu@ubuntu-VirtualBox:~$
```

Ubuntu ip allocation: 192.168.1.101



```
(kali@kali)~$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
    inet 192.168.2.100  netmask 255.255.255.0  broadcast 192.168.2.255
    inet6 fe80::ee9b:8888:db3e:c03  prefixlen 64  scopeid 0<20<link>
    ether 08:00:27:5a:87:bc  txqueuelen 1000  (Ethernet)
    RX packets 1  bytes 342 (342.0 B)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 23  bytes 3064 (2.9 KiB)
    TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
    inet 127.0.0.1  netmask 255.0.0.0
    inet6 ::1  prefixlen 128  scopeid 0<10<host>
    loop txqueuelen 1000  (Local Loopback)
    RX packets 8  bytes 480 (480.0 B)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 8  bytes 480 (480.0 B)
    TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0

(kali@kali)~$
```

Kali ip allocation: 192.168.2.100

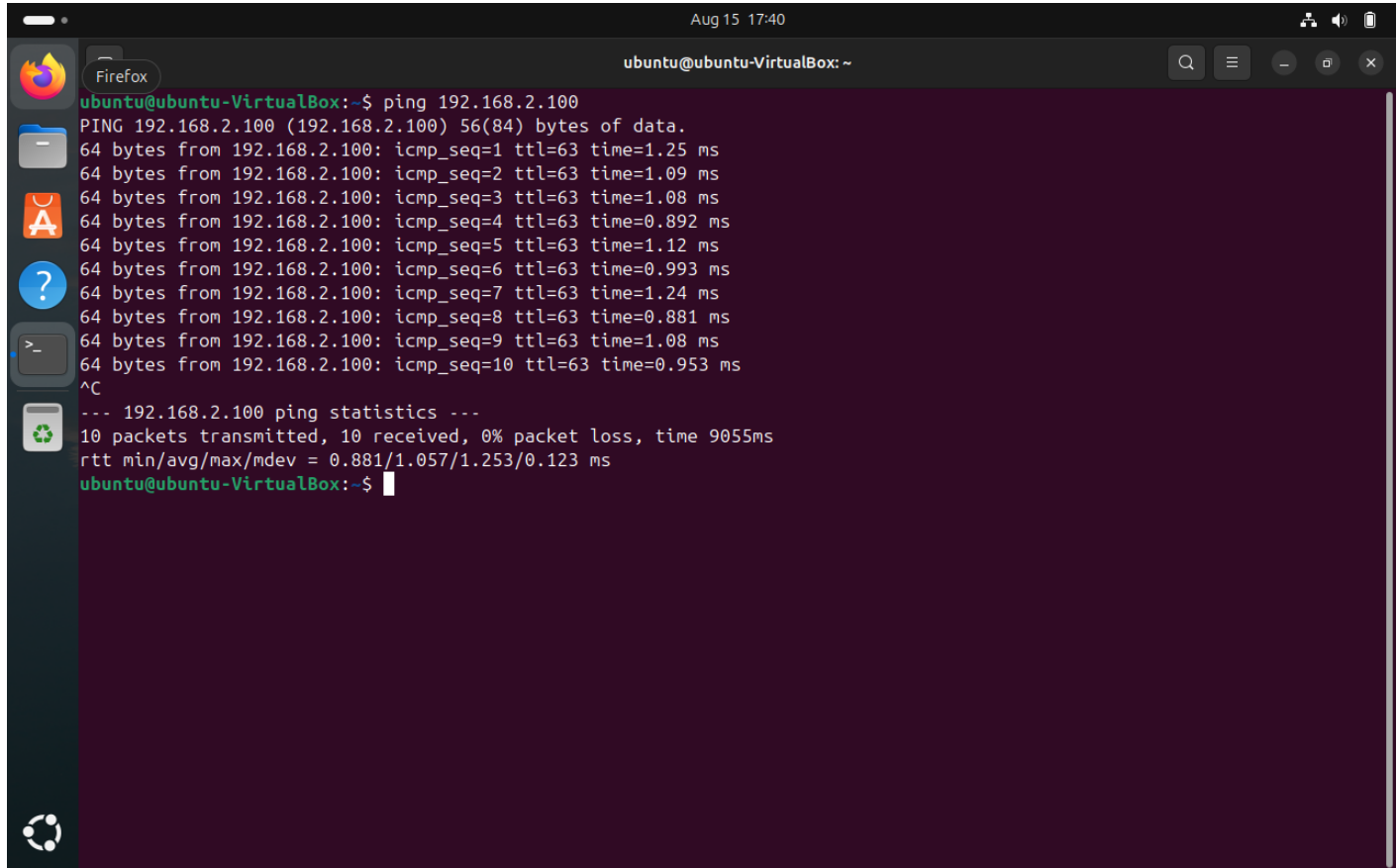
```

WAZUH Open Source Security Platform
https://wazuh.com
[wazuh-user@wazuh-server ~]$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.1.102 netmask 255.255.255.0 broadcast 192.168.1.255
    inet6 fe80::a00:27ff:fe01:25c prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:e1:02:5c txqueuelen 1000 (Ethernet)
    RX packets 690 bytes 818467 (799.2 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 700 bytes 51597 (50.3 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 3091 bytes 823922 (804.6 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 3091 bytes 823922 (804.6 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

```

Wazuh ip allocation: 192.168.1.102



The image shows a terminal window titled 'ubuntu@ubuntu-VirtualBox: ~' with a search icon and window controls. The terminal output shows a successful ping from the Ubuntu host to the Kali Linux IP address 192.168.2.100. The output includes 10 successful ping responses with varying times and a final statistics summary.

```
ubuntu@ubuntu-VirtualBox:~$ ping 192.168.2.100
PING 192.168.2.100 (192.168.2.100) 56(84) bytes of data:
64 bytes from 192.168.2.100: icmp_seq=1 ttl=63 time=1.25 ms
64 bytes from 192.168.2.100: icmp_seq=2 ttl=63 time=1.09 ms
64 bytes from 192.168.2.100: icmp_seq=3 ttl=63 time=1.08 ms
64 bytes from 192.168.2.100: icmp_seq=4 ttl=63 time=0.892 ms
64 bytes from 192.168.2.100: icmp_seq=5 ttl=63 time=1.12 ms
64 bytes from 192.168.2.100: icmp_seq=6 ttl=63 time=0.993 ms
64 bytes from 192.168.2.100: icmp_seq=7 ttl=63 time=1.24 ms
64 bytes from 192.168.2.100: icmp_seq=8 ttl=63 time=0.881 ms
64 bytes from 192.168.2.100: icmp_seq=9 ttl=63 time=1.08 ms
64 bytes from 192.168.2.100: icmp_seq=10 ttl=63 time=0.953 ms
^C
--- 192.168.2.100 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9055ms
rtt min/avg/max/mdev = 0.881/1.057/1.253/0.123 ms
ubuntu@ubuntu-VirtualBox:~$
```

Successful ping kali from ubuntu

```
Minimize all open windows and show the desktop
kali@kali: -
Session Actions Edit View Help

(kali@kali)-[~]
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.2.100 netmask 255.255.255.0 broadcast 192.168.2.255
    inet6 fe80::ee9b:8888:db3e:c03 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:5a:87:bc txqueuelen 1000 (Ethernet)
    RX packets 1 bytes 342 (342.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 23 bytes 3064 (2.9 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 8 bytes 480 (480.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8 bytes 480 (480.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

(kali@kali)-[~]
$ ping 192.168.1.101
PING 192.168.1.101 (192.168.1.101) 56(84) bytes of data:
64 bytes from 192.168.1.101: icmp_seq=1 ttl=63 time=0.841 ms
64 bytes from 192.168.1.101: icmp_seq=2 ttl=63 time=1.01 ms
64 bytes from 192.168.1.101: icmp_seq=3 ttl=63 time=1.02 ms
64 bytes from 192.168.1.101: icmp_seq=4 ttl=63 time=1.06 ms
64 bytes from 192.168.1.101: icmp_seq=5 ttl=63 time=1.04 ms
64 bytes from 192.168.1.101: icmp_seq=6 ttl=63 time=1.04 ms
64 bytes from 192.168.1.101: icmp_seq=7 ttl=63 time=0.885 ms
64 bytes from 192.168.1.101: icmp_seq=8 ttl=63 time=1.08 ms
64 bytes from 192.168.1.101: icmp_seq=9 ttl=63 time=0.948 ms
64 bytes from 192.168.1.101: icmp_seq=10 ttl=63 time=0.931 ms
64 bytes from 192.168.1.101: icmp_seq=11 ttl=63 time=0.850 ms
64 bytes from 192.168.1.101: icmp_seq=12 ttl=63 time=0.878 ms
^C
  — 192.168.1.101 ping statistics —
  12 packets transmitted, 12 received, 0% packet loss, time 11032ms
 rtt min/avg/max/mdev = 0.841/0.965/1.082/0.083 ms

(kali@kali)-[~]
$
```

Ping ubuntu from kali

Test port 22 for listen

```
valid_lft forever preferred_lft forever
ubuntu@ubuntu-VirtualBox:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.1.101 netmask 255.255.255.0 broadcast 192.168.1.255
    inet6 fe80::a00:27ff:face:96ec prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:ce:96:ec txqueuelen 1000 (Ethernet)
    RX packets 957 bytes 137703 (137.7 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1261 bytes 327477 (327.4 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 137 bytes 12413 (12.4 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 137 bytes 12413 (12.4 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

ubuntu@ubuntu-VirtualBox:~$ sudo systemctl enable ssh
[sudo] password for ubuntu:
Synchronizing state of ssh.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh
ubuntu@ubuntu-VirtualBox:~$ sudo ss -ltnp | grep :22
sudo: ss -ltnp: command not found
ubuntu@ubuntu-VirtualBox:~$ sudo ss -ltnp | grep :22
tcp    LISTEN 0      4096      0.0.0.0:*        0.0.0.0:*        users:(("sshd",pid=1323,fd=3),("systemd",pid=1,fd=409))
tcp    LISTEN 0      4096      [::]:22        [::]:*          users:(("sshd",pid=1323,fd=4),("systemd",pid=1,fd=410))

ubuntu@ubuntu-VirtualBox:~$
```

This is to confirm that port 22 is listen on ubuntu

Ubuntu agent is active on Wireshark

Minimize all open windows and show the desktop | 1 2 3 4 | Wazuh

192.168.1.102/app/endpoints-summary#/agents-preview/ 90% ☆

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W. Endpoints API default a

AGENTS BY STATUS

- Active (1)
- Disconnected (1)
- Pending (0)
- Never connected (0)

TOP 5 OS

- kali (1)
- ubuntu (1)

TOP 5 GROUPS

- default (2)

Agents (1)

Deploy new agent Refresh Export formatted More

status=active WQL

ID ↑	Name	IP address	Group(s)	Operating system	Cluster node	Version	Status	Actions
004	Blessagent2	192.168.1.101	default	Ubuntu 24.04.4 LTS	node01	v4.14.7	active	👁️ ⋮

Rows per page: 10 < 1 >

pfSense.home.arpa - Stat x pfSense.home.arpa - Stat x Problem loading page x Screenshot_2026-08-15_17_ x +

192.168.1.1 110% ☆

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UAT Crypto: No

Hardware crypto	Inactive
Kernel PTI	Disabled
MDS Mitigation	Inactive
Uptime	12 Hours 48 Minutes 39 Seconds
Current date/time	Wed Aug 19 21:14:48 GMT 2026
DNS server(s)	127.0.0.1 ::1 194.168.4.100 194.168.8.100
Last config change	Wed Aug 19 20:43:04 GMT 2026
State table size	0% (46/96000) Show states
MBUF Usage	1% (5566/1000000)
Load average	0.85, 0.75, 0.72
CPU usage	2%
Memory usage	33% of 968 MiB

FAQ Netgate

- Netgate Professional Services
- Visit Netgate.com

If you decide to purchase a Netgate Global TAC Support subscription, you **MUST** have your **Netgate Device ID (NDI)** from your firewall in order to validate support for this unit. Write down your NDI and store it in a safe place. You can purchase a TAC support plan [here](#).

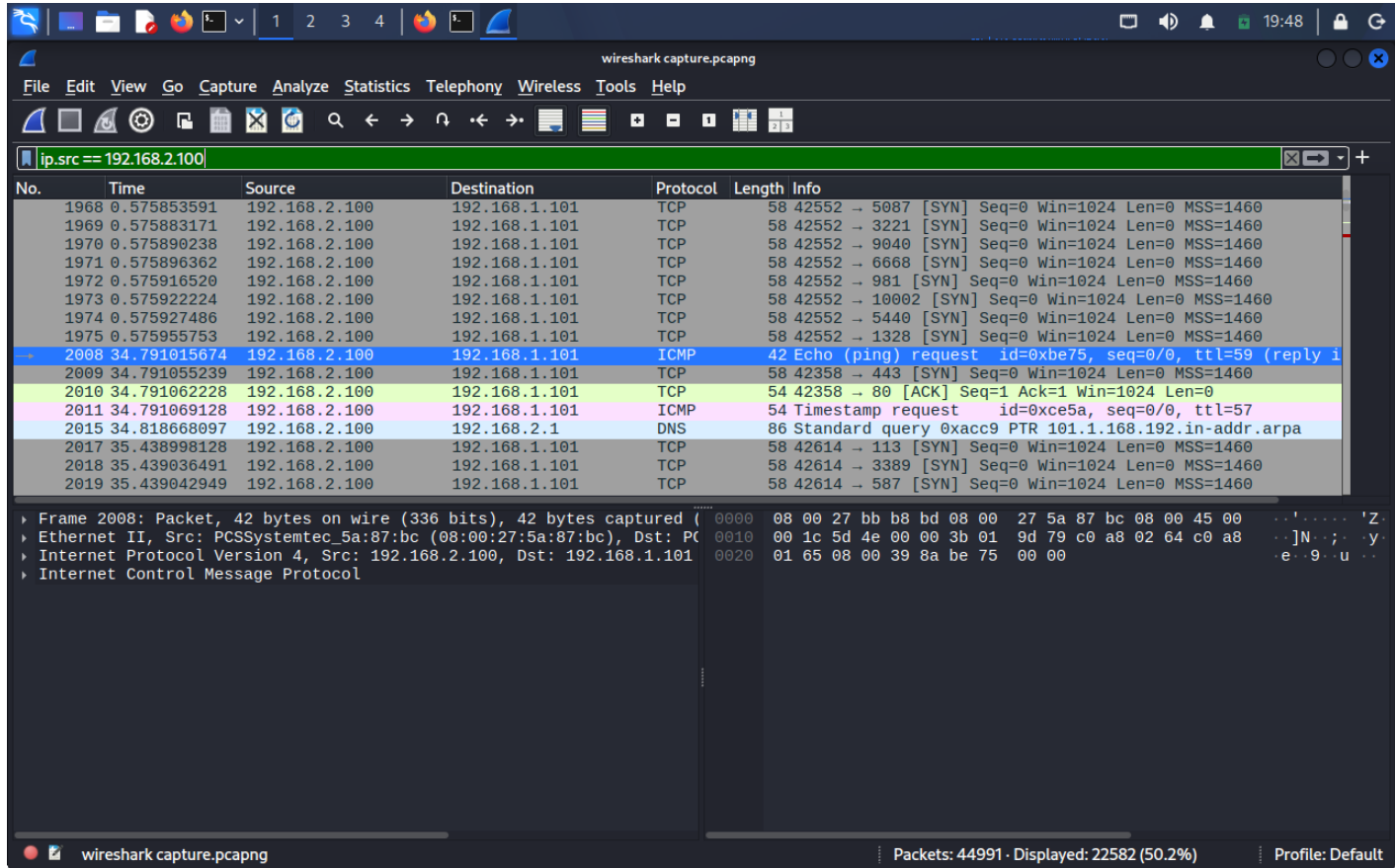
Interfaces

WAN	↑	1000baseT <full-duplex>	10.0.2.15 fd17:625c:f037:2:a00:27ff:fe00:d488
LAN	↑	1000baseT <full-duplex>	192.168.1.1
DMZ	↑	1000baseT <full-duplex>	192.168.2.1

Snort Alerts

Interface/Time	Src/Dst Address	Description
WAN	[fe80::2]	ICMP Packet Detected
Aug 19 21:14...	[fd17:625c:f037:2:a00:2...	
WAN	[fe80::2]	ICMP Packet Detected

Pfsense firewall is up. WAN LAN DMZ interfaces are active.



All traffic to Kali Ip on Wireshark

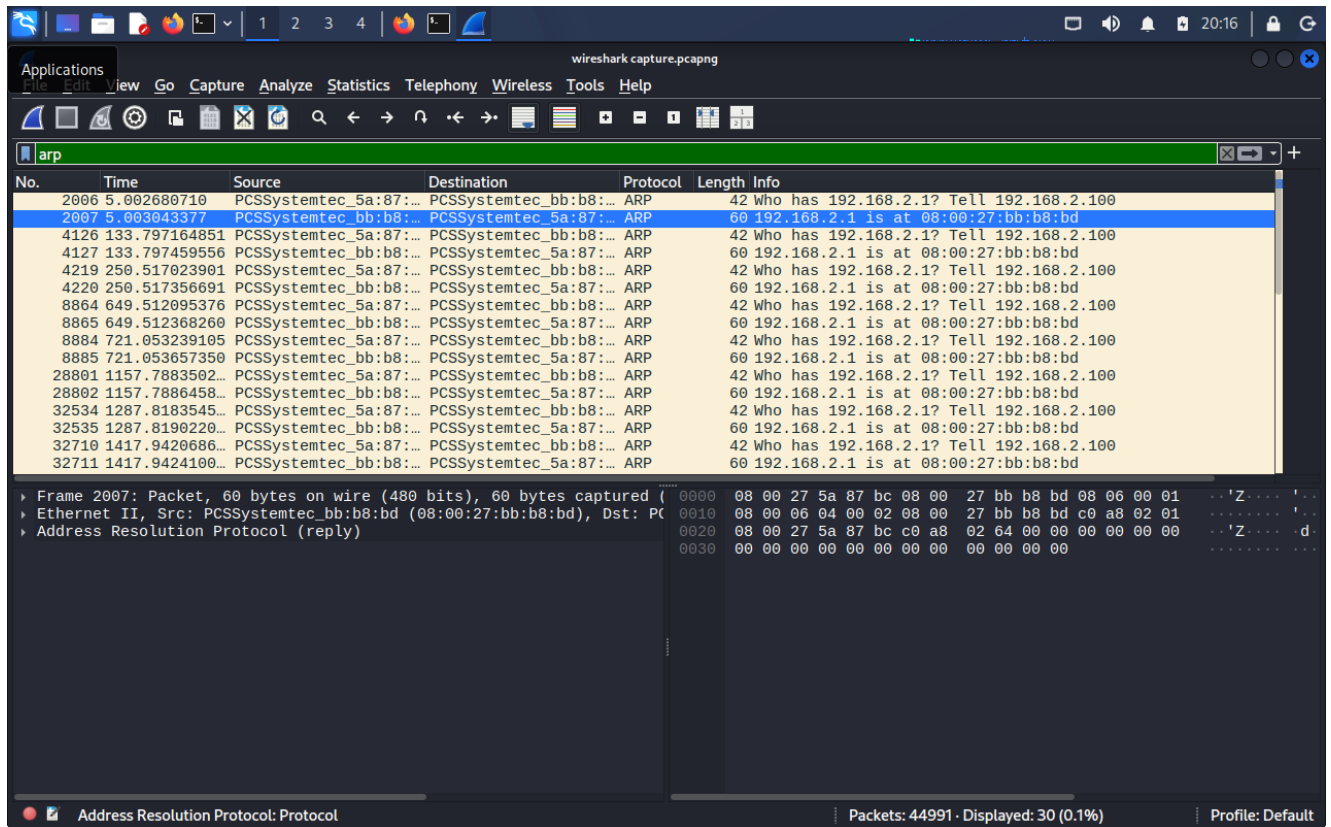
Wireshark network traffic capture showing multiple failed SSH attempts to port 22. The packet list shows several RST (Reset) packets from 192.168.2.100 to 192.168.1.102. The packet details pane shows the structure of a failed SYN packet.

No.	Time	Source	Destination	Protocol	Length	Info
56	226.219472260	192.168.2.100	192.168.1.102	TCP	58	37138 → 22 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
66	226.220573834	192.168.1.102	192.168.2.100	TCP	60	22 → 37138 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
68	226.220623763	192.168.2.100	192.168.1.102	TCP	54	37138 → 22 [RST] Seq=1 Win=0 Len=0
2077	308.805238281	192.168.2.100	192.168.1.102	TCP	58	36795 → 22 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
2078	308.806061455	192.168.1.102	192.168.2.100	TCP	60	22 → 36795 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
2079	308.806088269	192.168.2.100	192.168.1.102	TCP	54	36795 → 22 [RST] Seq=1 Win=0 Len=0
2080	308.920436714	192.168.2.100	192.168.1.102	TCP	74	43342 → 22 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PER
2081	308.921226758	192.168.1.102	192.168.2.100	TCP	74	22 → 43342 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460
2082	308.921275654	192.168.2.100	192.168.1.102	TCP	66	43342 → 22 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=33698
2083	308.983752383	192.168.1.102	192.168.2.100	SSH	87	Server: Protocol (SSH-2.0-OpenSSH_8.7)
2084	308.983795134	192.168.2.100	192.168.1.102	TCP	66	43342 → 22 [ACK] Seq=1 Ack=22 Win=64256 Len=0 TSval=33698
2085	308.989135489	192.168.2.100	192.168.1.102	TCP	66	43342 → 22 [FIN, ACK] Seq=1 Ack=22 Win=64256 Len=0 TSval=33698
2086	308.992980579	192.168.1.102	192.168.2.100	TCP	66	22 → 43342 [FIN, ACK] Seq=22 Ack=2 Win=65280 Len=0 TSval=33698
2087	308.993027009	192.168.2.100	192.168.1.102	TCP	66	43342 → 22 [ACK] Seq=2 Ack=23 Win=64256 Len=0 TSval=33698
2210	399.795017486	192.168.2.100	192.168.1.101	TCP	74	33104 → 22 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PER
2211	399.795705620	192.168.1.101	192.168.2.100	TCP	74	22 → 33104 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460

Frame 56: Packet, 58 bytes on wire (464 bits), 58 bytes captured (464 bits) on interface eth0
 Ethernet II, Src: PCSSystemtec_5a:87:bc (08:00:27:5a:87:bc), Dst: PCSSystemtec_5a:87:bc (08:00:27:5a:87:bc)
 Internet Protocol Version 4, Src: 192.168.2.100, Dst: 192.168.1.102
 Transmission Control Protocol, Src Port: 37138, Dst Port: 22, Seq: 0, Win: 1024, Len: 0

Wireshark: port 22 shows kali is repeatedly attempting to reach SSH particularly ubuntu with a mixture of successful 3 way handshakes.

ARP FLOODING



The image shows a Wireshark capture of network traffic, specifically ARP requests and replies. The capture is filtered for 'arp' and shows a list of packets in the main pane. The packet list pane shows 32 packets, all of which are ARP requests from PCSSystemtec_5a:87:... to PCSSystemtec_bb:b8:... . The packet details pane shows the selected packet (No. 2007) and its details: Ethernet II, Src: PCSSystemtec_bb:b8:bd (08:00:27:bb:b8:bd), Dst: PCSSystemtec_5a:87:... (08:00:27:5a:87:bc), and Address Resolution Protocol (reply). The packet bytes pane shows the raw data of the packet.

No.	Time	Source	Destination	Protocol	Length	Info
2006	5.002680710	PCSSystemtec_5a:87:...	PCSSystemtec_bb:b8:...	ARP	42	Who has 192.168.2.1? Tell 192.168.2.100
2007	5.003043377	PCSSystemtec_bb:b8:...	PCSSystemtec_5a:87:...	ARP	60	192.168.2.1 is at 08:00:27:bb:b8:bd
4126	133.797164851	PCSSystemtec_5a:87:...	PCSSystemtec_bb:b8:...	ARP	42	Who has 192.168.2.1? Tell 192.168.2.100
4127	133.797459556	PCSSystemtec_bb:b8:...	PCSSystemtec_5a:87:...	ARP	60	192.168.2.1 is at 08:00:27:bb:b8:bd
4219	250.517023901	PCSSystemtec_5a:87:...	PCSSystemtec_bb:b8:...	ARP	42	Who has 192.168.2.1? Tell 192.168.2.100
4220	250.517356691	PCSSystemtec_bb:b8:...	PCSSystemtec_5a:87:...	ARP	60	192.168.2.1 is at 08:00:27:bb:b8:bd
8864	649.512095376	PCSSystemtec_5a:87:...	PCSSystemtec_bb:b8:...	ARP	42	Who has 192.168.2.1? Tell 192.168.2.100
8865	649.512368260	PCSSystemtec_bb:b8:...	PCSSystemtec_5a:87:...	ARP	60	192.168.2.1 is at 08:00:27:bb:b8:bd
8884	721.053239105	PCSSystemtec_5a:87:...	PCSSystemtec_bb:b8:...	ARP	42	Who has 192.168.2.1? Tell 192.168.2.100
8885	721.053657350	PCSSystemtec_bb:b8:...	PCSSystemtec_5a:87:...	ARP	60	192.168.2.1 is at 08:00:27:bb:b8:bd
28801	1157.7883502...	PCSSystemtec_5a:87:...	PCSSystemtec_bb:b8:...	ARP	42	Who has 192.168.2.1? Tell 192.168.2.100
28802	1157.7886458...	PCSSystemtec_bb:b8:...	PCSSystemtec_5a:87:...	ARP	60	192.168.2.1 is at 08:00:27:bb:b8:bd
32534	1287.8183545...	PCSSystemtec_5a:87:...	PCSSystemtec_bb:b8:...	ARP	42	Who has 192.168.2.1? Tell 192.168.2.100
32535	1287.8190220...	PCSSystemtec_bb:b8:...	PCSSystemtec_5a:87:...	ARP	60	192.168.2.1 is at 08:00:27:bb:b8:bd
32710	1417.9420686...	PCSSystemtec_5a:87:...	PCSSystemtec_bb:b8:...	ARP	42	Who has 192.168.2.1? Tell 192.168.2.100
32711	1417.9424100...	PCSSystemtec_bb:b8:...	PCSSystemtec_5a:87:...	ARP	60	192.168.2.1 is at 08:00:27:bb:b8:bd

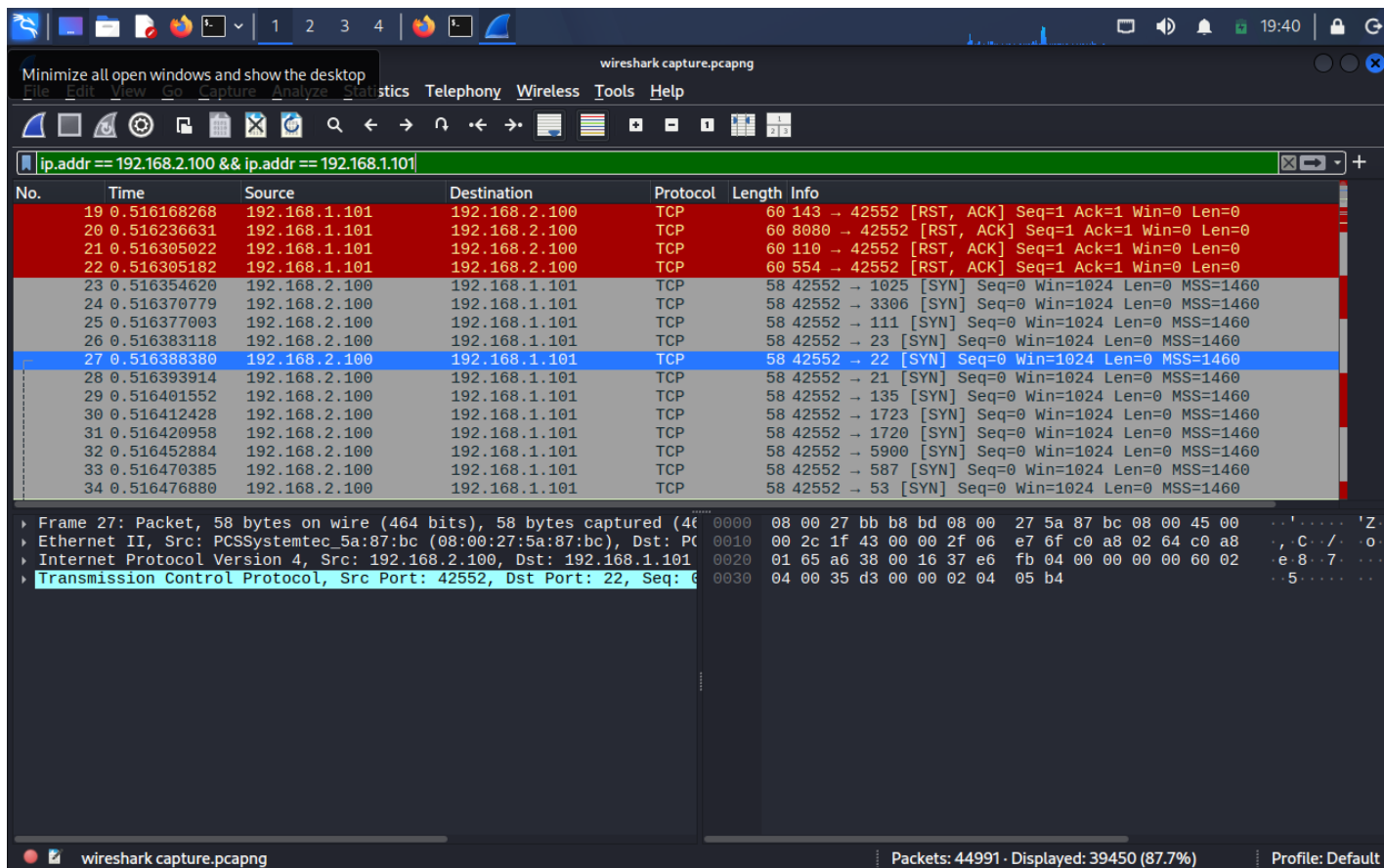
Frame 2007: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface 0
Ethernet II, Src: PCSSystemtec_bb:b8:bd (08:00:27:bb:b8:bd), Dst: PCSSystemtec_5a:87:... (08:00:27:5a:87:bc)
Address Resolution Protocol (reply)

Address Resolution Protocol: Protocol

Packets: 44991 · Displayed: 30 (0.1%)

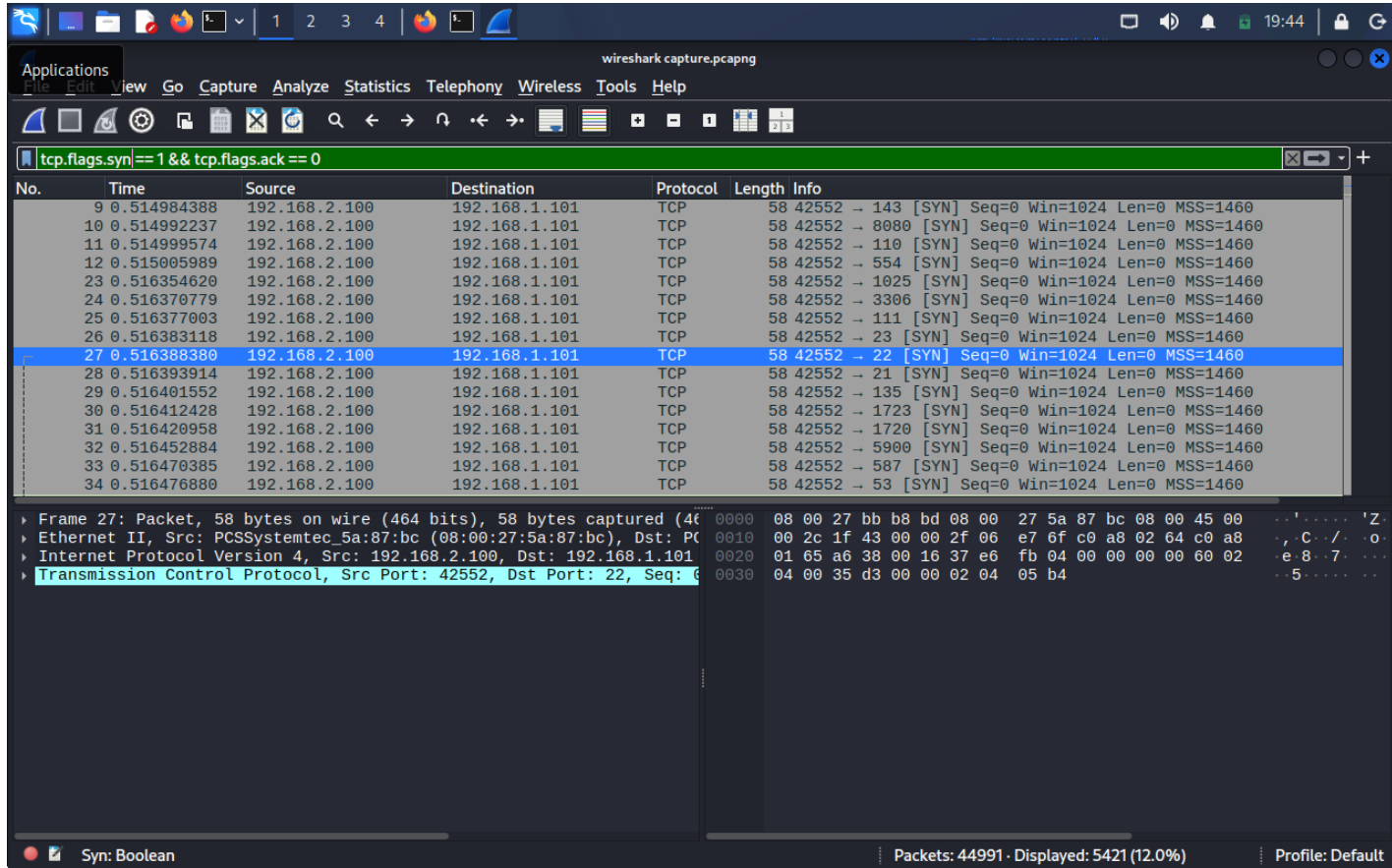
Profile: Default

The Wireshark evidence indicates that 192.168.2.100 is performing a TCP SYN-based port scan against 192.168.1.101. The activity should be classified as network reconnaissance / port scanning rather than normal TCP communication. During a SYN scan, the scanner sends SYN packets but generally does not complete normal TCP connections.



This capture strongly shows a **TCP port scan** from **192.168.2.100 (Kali)** against **192.168.2.101 (Ubuntu)**. A SYN is the first packet used to establish a TCP connection. Here, however, kali is rapidly sending SYN packets to **many different ports**, which means reconnaissance/port scanning is going on. RST, ACK generally means the target received the connection attempt but **there isn't a listening TCP service on that port**

- **One source IP:** 192.168.2.100
- **One target IP:** 192.168.2.101
- **Many different destination ports**
- **Very short time interval**



Wireshark filter used:

SSH traffic (`tcp.port == 22`)

Traffic between kali and ubuntu (`ip.addr == 192.168.2.100 && ip.addr == 192.168.1.101`)

All traffic to ubuntu (`ip.dst == 192.168.1.101`)

All traffic to Kali (`ip.sre == 192.168.2.100`)

ICMP echo request (`icmp.type == 8`)

TCP SYN scan detection (`tcp.flags.syn == 1 && tcp.flags.ack == 0`)

```
kali@kali: ~  
Session Actions Edit View Help  
64 bytes from 192.168.2.1: icmp_seq=15 ttl=64 time=0.568 ms  
64 bytes from 192.168.2.1: icmp_seq=16 ttl=64 time=0.508 ms  
64 bytes from 192.168.2.1: icmp_seq=17 ttl=64 time=0.564 ms  
64 bytes from 192.168.2.1: icmp_seq=18 ttl=64 time=0.735 ms  
64 bytes from 192.168.2.1: icmp_seq=19 ttl=64 time=0.422 ms  
64 bytes from 192.168.2.1: icmp_seq=20 ttl=64 time=0.949 ms  
64 bytes from 192.168.2.1: icmp_seq=21 ttl=64 time=0.661 ms  
64 bytes from 192.168.2.1: icmp_seq=22 ttl=64 time=0.798 ms  
64 bytes from 192.168.2.1: icmp_seq=23 ttl=64 time=0.669 ms  
64 bytes from 192.168.2.1: icmp_seq=24 ttl=64 time=0.530 ms  
^C  
— 192.168.2.1 ping statistics —  
24 packets transmitted, 24 received, 0% packet loss, time 26164ms  
rtt min/avg/max/mdev = 0.422/0.620/0.949/0.129 ms  
  
(kali@kali)-[~]  
$ nmap -Pn 192.168.1.101  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-17 20:31 +0100  
Nmap scan report for 192.168.1.101  
Host is up (0.0016s latency).  
Not shown: 997 closed tcp ports (reset)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
22/tcp    filtered ssh  
80/tcp    open  http  
  
Nmap done: 1 IP address (1 host up) scanned in 1.84 seconds  
  
(kali@kali)-[~]  
$ nmap -Pn 192.168.1.101  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-17 20:33 +0100  
Nmap scan report for 192.168.1.101  
Host is up (0.0013s latency).  
Not shown: 997 closed tcp ports (reset)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
22/tcp    filtered ssh  
80/tcp    open  http  
  
Nmap done: 1 IP address (1 host up) scanned in 0.76 seconds  
  
(kali@kali)-[~]  
$
```

Port scanning commands

```
kali@kali: ~  
Session Actions Edit View Help  
ls  
Desktop Documents Downloads Music password.txt Pictures Projects Public Templates Videos  
kali@kali: ~  
$ sudo nmap -sS -p 1-1000 192.168.1.102  
[sudo] password for kali:  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-17 20:55 +0100  
Nmap scan report for 192.168.1.102  
Host is up (0.0029s latency).  
Not shown: 997 closed tcp ports (reset)  
PORT      STATE SERVICE  
22/tcp    open  ssh  
443/tcp   open  https  
514/tcp   open  shell  
Nmap done: 1 IP address (1 host up) scanned in 0.72 seconds  
kali@kali: ~  
$ sudo nmap -sV -p 22 192.168.1.102  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-17 20:56 +0100  
Nmap scan report for 192.168.1.102  
Host is up (0.00071s latency).  
on wire (1088 bits), 281 bytes captured on interface eth0  
PORT      STATE SERVICE VERSION  
22/tcp    open  ssh      OpenSSH 8.7 (protocol 2.0)  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 1.06 seconds  
kali@kali: ~  
$
```

eth0: live capture in progress

Packets: 2091

Profile: Default

```
kali@kali: ~  
Session Actions Edit View Help  
Nmap done: 1 IP address (1 host up) scanned in 1.06 seconds  
  
(kali@kali)-[~]  
$ hydra -l ubuntu -P password.txt ssh://192.168.1.101  
Hydra v9.7 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).  
  
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-08-17 20:58:22  
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4  
[DATA] max 6 tasks per 1 server, overall 6 tasks, 6 login tries (l:p:6), ~1 try per task  
[DATA] attacking ssh://192.168.1.101:22/  
[22][ssh] host: 192.168.1.101 login: ubuntu password: ubuntu  
1 of 1 target successfully completed, 1 valid password found  
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-08-17 20:58:26  
  
(kali@kali)-[~]  
$ ssh ubuntu@192.168.1.101  
ubuntu@192.168.1.101's password:  
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 7.0.0-28-generic x86_64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:        https://ubuntu.com/pro  
  
Expanded Security Maintenance for Applications is not enabled.  
0 updates can be applied immediately.  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
The list of available updates is more than a week old.  
To check for new updates run: sudo apt update  
Last login: Fri Aug 14 11:26:56 2026 from 192.168.1.100  
ubuntu@ubuntu-VirtualBox:~$ sudo su  
[sudo] password for ubuntu:   
Sorry, try again.  
[sudo] password for ubuntu:   
root@ubuntu-VirtualBox:/home/ubuntu#  
root@ubuntu-VirtualBox:/home/ubuntu# exit  
exit  
ubuntu@ubuntu-VirtualBox:~$ exit  
exit
```

Hydra password-guessing attack

```
kali@kali: ~  
Session Actions Edit View Help  
└─$ nmap -Pn 192.168.1.101  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-17 21:03 +0100  
Nmap scan report for 192.168.1.101  
Host is up (0.0012s latency).  
Not shown: 997 closed tcp ports (reset)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
80/tcp    open  http  
Nmap done: 1 IP address (1 host up) scanned in 0.70 seconds  
(kali@kali)-[~]  
└─$ nikto -h 192.168.1.101  
- Nikto v2.6.0  
+ Target IP: 192.168.1.101  
+ Target Hostname: 192.168.1.101  
+ Target Port: 80  
+ Platform: Linux/Unix  
+ Start Time: 2026-08-17 21:05:21 (GMT1)  
+ Server: Apache/2.4.58 (Ubuntu)  
+ ERROR: Failed to check for updates:  
+ No CGI Directories found (use '-C all' to force check all possible dirs). CGI tests skipped.  
+ [999984] /: Server may leak inodes via ETags, header found with file /, inode: 29af, size: 6576d9cc2c76b, mtime: gzip. See: https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2003-1418  
+ [013587] /: Suggested security header missing: permissions-policy. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Permissions-Policy  
+ [013587] /: Suggested security header missing: referrer-policy. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Referrer-Policy  
+ [013587] /: Suggested security header missing: strict-transport-security. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Strict-Transport-Security  
+ [013587] /: Suggested security header missing: x-content-type-options. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/X-Content-Type-Options  
+ [013587] /: Suggested security header missing: content-security-policy. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/CSP  
+ [600050] Apache/2.4.58 appears to be outdated (current is at least 2.4.66).  
+ [999990] OPTIONS: Allowed HTTP Methods: GET, POST, OPTIONS, HEAD .  
^C  
(kali@kali)-[~]  
└─$ nano passwordlist.txt  
(kali@kali)-[~]  
└─$  
Packets: 23564 Dropped: 0 (0.0%) Profile: Default
```

Nikto command

192.168.1.1/snort/snort_alerts.php										Guessing Attack
2026-08-17 20:57:01	2	TCP	Attempted Information Leak	192.168.1.101	80	192.168.2.100	59662	1:2009885	ET SCAN Unusually Fast 404 Error Messages (Page Not Found), Possible Web Application Scan/Directory Guessing Attack	
2026-08-17 20:49:53	2	TCP	Attempted Information Leak	192.168.2.100	42108	192.168.1.101	22	1:2003068	ET SCAN Potential SSH Scan OUTBOUND	
2026-08-17 20:49:53	1	TCP	Web Application Attack	192.168.2.100	36852	192.168.1.101	80	1:2024364	ET SCAN Possible Nmap User-Agent Observed	
2026-08-17 20:49:53	1	TCP	Web Application Attack	192.168.2.100	36852	192.168.1.101	80	1:2009358	ET SCAN Nmap Scripting Engine User-Agent Detected (Nmap Scripting Engine)	
2026-08-17 20:49:52	1	TCP	Web Application Attack	192.168.2.100	36844	192.168.1.101	80	1:2024364	ET SCAN Possible Nmap User-Agent Observed	
2026-08-17 20:49:52	1	TCP	Web Application	192.168.2.100	36844	192.168.1.101	80	1:2009358	ET SCAN Nmap Scripting Engine	

The screenshot shows **Snort IDS alerts in pfSense** detecting suspicious network activity. The host 192.168.2.100 is scanning 192.168.1.101 on **HTTP (port 80)** and **SSH (port 22)**. Alerts indicate possible **Nmap scanning, web application enumeration, and directory guessing**.

System Firewall DHCP Authentication IPsec PPP PPPoE/L2TP Server OpenVPN NTP Packages Settings

Normal View Dynamic View Summary View

Last 500 Firewall Log Entries. (Maximum 500) Pause

Action	Time	Interface	Source	Destination	Protocol
✓	Aug 17 18:57:56	DMZ	192.168.2.100:43014	192.168.1.102:443	TCP:S
✓	Aug 17 18:57:51	DMZ	192.168.2.100:40600	199.232.53.91:443	TCP:S
✓	Aug 17 18:57:51	DMZ	192.168.2.100:42919	192.168.2.1:53	UDP
✓	Aug 17 18:57:51	DMZ	192.168.2.100:59178	34.49.51.44:443	TCP:S
✓	Aug 17 18:57:47	DMZ	192.168.2.100:40586	199.232.53.91:443	TCP:S
✓	Aug 17 18:57:47	DMZ	192.168.2.100:40578	199.232.53.91:443	TCP:S
✓	Aug 17 18:57:47	DMZ	192.168.2.100:46186	192.168.2.1:53	UDP
✓	Aug 17 18:57:46	DMZ	192.168.2.100:47904	151.101.129.91:443	TCP:S
✓	Aug 17 18:57:46	DMZ	192.168.2.100:36689	192.168.2.1:53	UDP
✓	Aug 17 18:57:46	DMZ	192.168.2.100:43015	192.168.2.1:53	UDP
✓	Aug 17 18:57:45	DMZ	192.168.2.100:55026	192.168.1.102:443	TCP:S
✓	Aug 17 18:57:43	DMZ	192.168.2.100:55020	192.168.1.102:443	TCP:S

The same source IP, different source ports, different destinations, and different ports. This means there's a Network Reconnaissance / Scanning.

										Scripting Engine)
2026-08-17 20:49:49	⚠	2	UDP	Attempted Information Leak	192.168.2.100 54979	192.168.1.101 44328	1:2018489	+	×	ET SCAN NMAP OS Detection Probe
2026-08-17 20:49:46	⚠	2	UDP	Attempted Information Leak	192.168.2.100 54723	192.168.1.101 44328	1:2018489	+	×	ET SCAN NMAP OS Detection Probe
2026-08-17 20:49:38	⚠	2	TCP	Attempted Information Leak	192.168.2.100 43605	192.168.1.101 5800	1:2002910	+	×	ET SCAN Potential VNC Scan 5800-5820
2026-08-17 20:49:38	⚠	2	TCP	Potentially Bad Traffic	192.168.2.100 43605	192.168.1.101 5432	1:2010939	+	×	ET SCAN Suspicious inbound to PostgreSQL port 5432
2026-08-17 20:49:38	⚠	2	TCP	Potentially Bad Traffic	192.168.2.100 43605	192.168.1.101 1433	1:2010935	+	×	ET SCAN Suspicious inbound to MSSQL port 1433
2026-08-17 20:49:38	⚠	2	TCP	Potentially Bad Traffic	192.168.2.100 43605	192.168.1.101 1521	1:2010936	+	×	ET SCAN Suspicious inbound to Oracle SQL port 1521
2026-08-17 20:49:38	⚠	2	TCP	Potentially Bad Traffic	192.168.2.100 43605	192.168.1.101 3306	1:2010937	+	×	ET SCAN Suspicious

IDS(snort): This is reporting a major breakdown of the nature of the traffic.

Aug 21 13:43

Wazuh x pfSense.home.arpa - Serv x

Not Secure 192.168.1.1/snort/snort_alerts.php?instance=1

2026-08-17 20:42:23	2	TCP	Attempted Information Leak	192.168.1.101 80	192.168.2.100 39596	1:2009885	ET SCAN Unusually Fast 404 Error Messages (Page Not Found), Possible Web Application Scan/Directory Guessing Attack
2026-08-17 20:42:23	2	TCP	Attempted Information Leak	192.168.1.101 80	192.168.2.100 39596	1:2009885	ET SCAN Unusually Fast 404 Error Messages (Page Not Found), Possible Web Application Scan/Directory Guessing Attack
2026-08-17 20:42:23	2	TCP	Attempted Information Leak	192.168.1.101 80	192.168.2.100 39596	1:2009885	ET SCAN Unusually Fast 404 Error Messages (Page Not Found), Possible Web Application Scan/Directory Guessing Attack
2026-08-17 20:42:23	2	TCP	Attempted Information Leak	192.168.1.101 80	192.168.2.100 39586	1:2009885	ET SCAN Unusually Fast 404 Error Messages (Page Not Found), Possible Web Application Scan/Directory Guessing Attack
2026-08-17 20:42:22	2	TCP	Attempted Information Leak	192.168.1.101 80	192.168.2.100 39586	1:2009885	ET SCAN Unusually Fast 404 Error Messages (Page Not Found), Possible Web Application Scan/Directory Guessing Attack
2026-08-17 20:42:22	2	TCP	Attempted Information Leak	192.168.1.101 80	192.168.2.100 39586	1:2009885	ET SCAN Unusually Fast 404 Error Messages (Page Not Found), Possible Web Application Scan/Directory Guessing Attack
2026-08-17 20:42:22	2	TCP	Attempted Information Leak	192.168.1.101 80	192.168.2.100 39586	1:2009885	ET SCAN Unusually Fast 404 Error Messages (Page Not Found), Possible Web Application Scan/Directory Guessing Attack
2026-08-17 20:42:22	2	TCP	Attempted Information Leak	192.168.1.101 80	192.168.2.100 39574	1:2009885	ET SCAN Unusually Fast 404 Error Messages (Page Not Found), Possible Web Application Scan/Directory Guessing Attack

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Ubuntu responding to kali on port 80.

```
Minimize all open windows and show the desktop
kali@kali: ~
$ hydra -l ubuntu -P password.txt ssh://192.168.1.101
Hydra v9.7 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-08-18 20:04:50
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4
[DATA] max 6 tasks per 1 server, overall 6 tasks, 6 login tries (l:1/p:6), ~1 try per task
[DATA] attacking ssh://192.168.1.101:22/
[22][ssh] host: 192.168.1.101 login: ubuntu password: ubuntu
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-08-18 20:04:55

(kali@kali)-[~]
$ hydra -l ubuntu -P password.txt ssh://192.168.1.101
Hydra v9.7 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

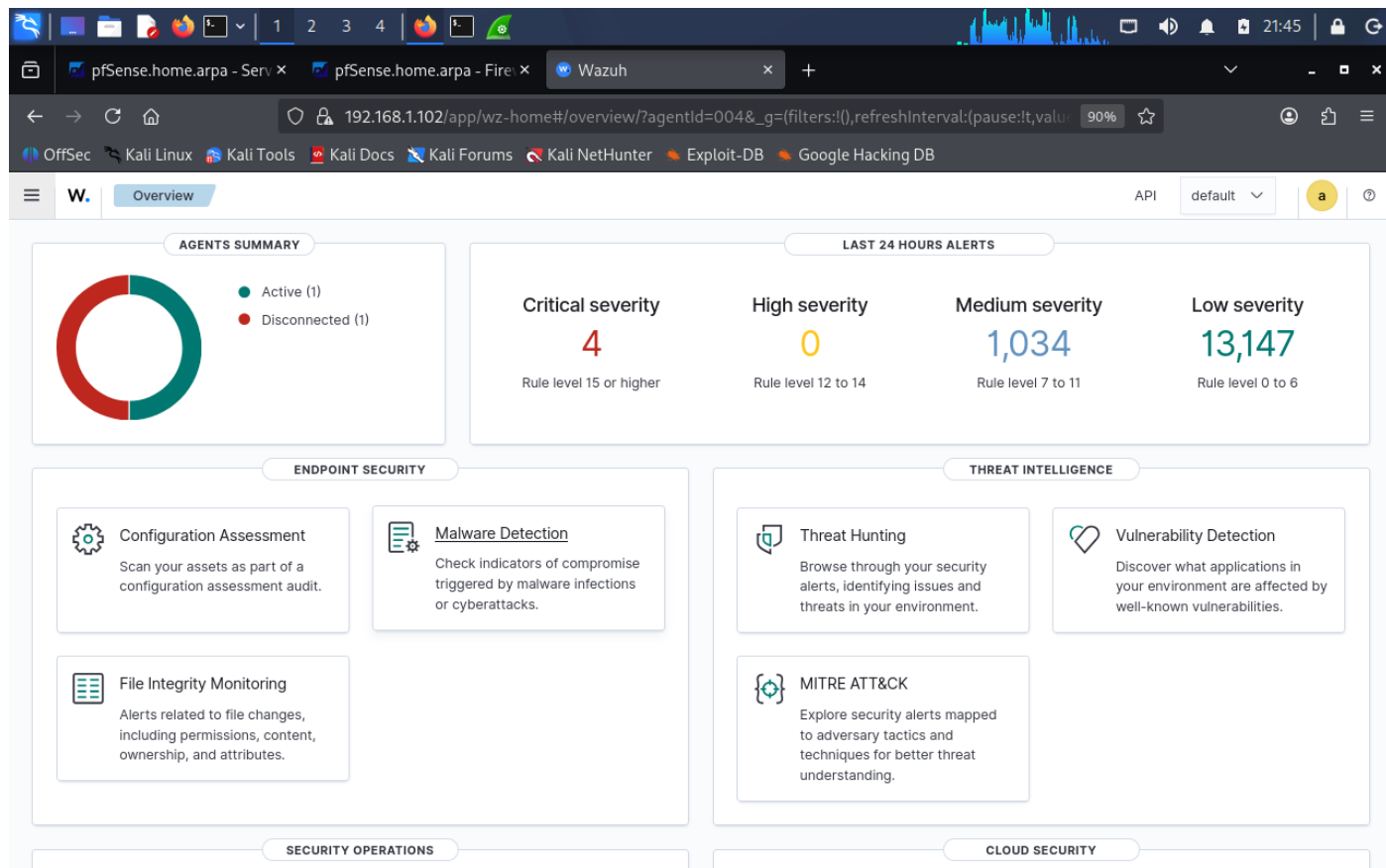
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-08-18 20:05:43
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4
[DATA] max 6 tasks per 1 server, overall 6 tasks, 6 login tries (l:1/p:6), ~1 try per task
[DATA] attacking ssh://192.168.1.101:22/
1 of 1 target completed, 0 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-08-18 20:05:49

(kali@kali)-[~]
$ hydra -l ubuntu -P password.txt ftp://192.168.1.101
Hydra v9.7 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

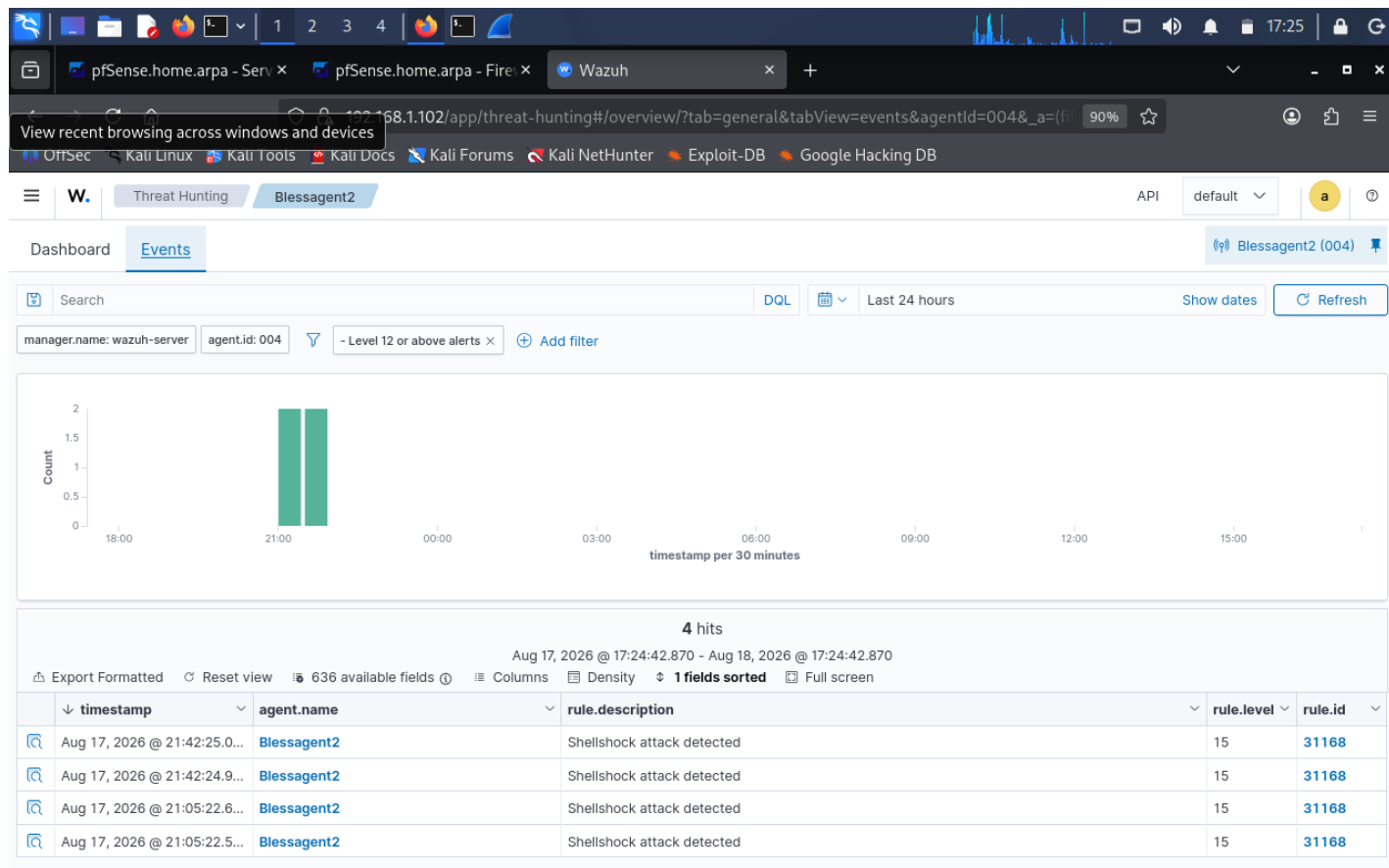
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-08-18 20:07:48
[DATA] max 6 tasks per 1 server, overall 6 tasks, 6 login tries (l:1/p:6), ~1 try per task
[DATA] attacking ftp://192.168.1.101:21/
[21][ftp] host: 192.168.1.101 login: ubuntu password: ubuntu
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-08-18 20:07:52

(kali@kali)-[~]
$
```

Simulation of attacks on unencrypted (FTP) and encrypted (SSH) endpoints:



This dashboard shows that the attacks are critical and also a medium severity



SIEM Critical alert which indicates a potential Shellshock attack against Ubuntu-hosted service.

Wazuh

192.168.1.102/app/threat-hunting#/overview/?tab=general&tabView=events&agentId=004&a=(f)

OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB

Threat Hunting Blessagent2

API default

28 hits

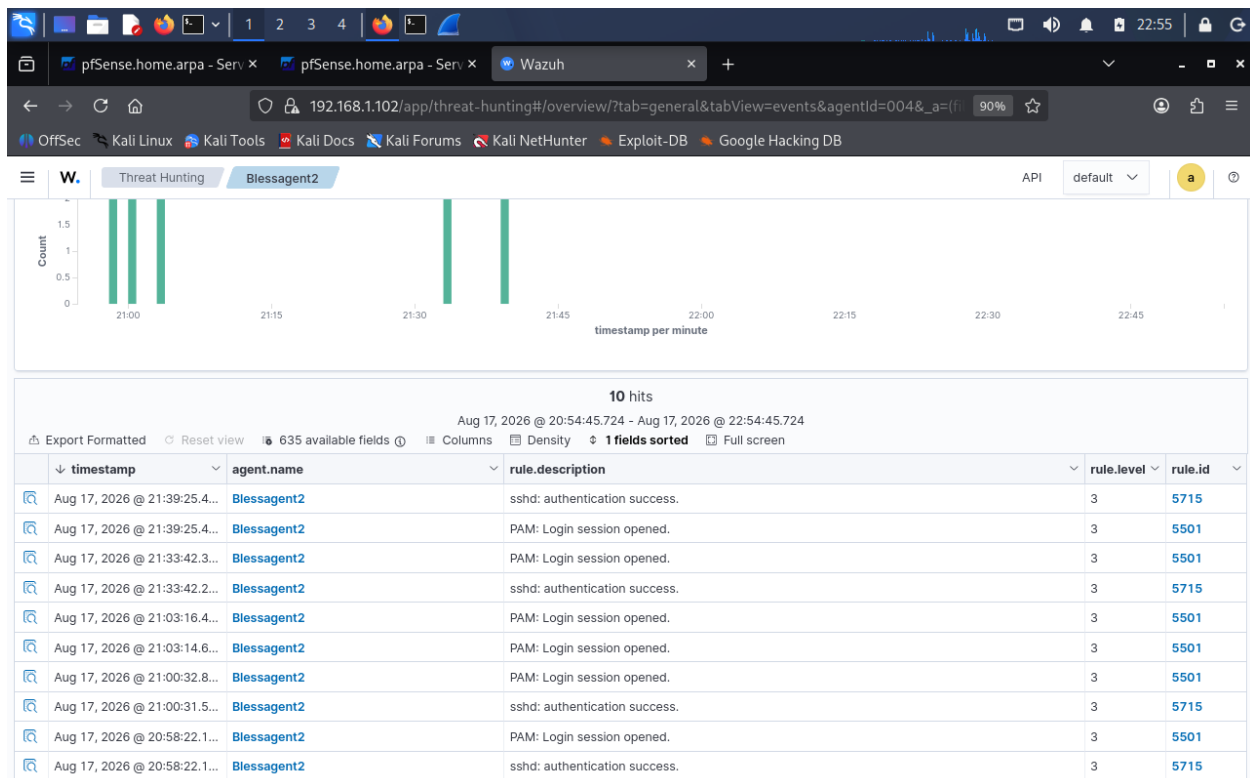
Aug 17, 2026 @ 20:33:09.478 - Aug 17, 2026 @ 22:33:09.478

Export Formatted Reset view 635 available fields Columns Density 1 fields sorted Full screen

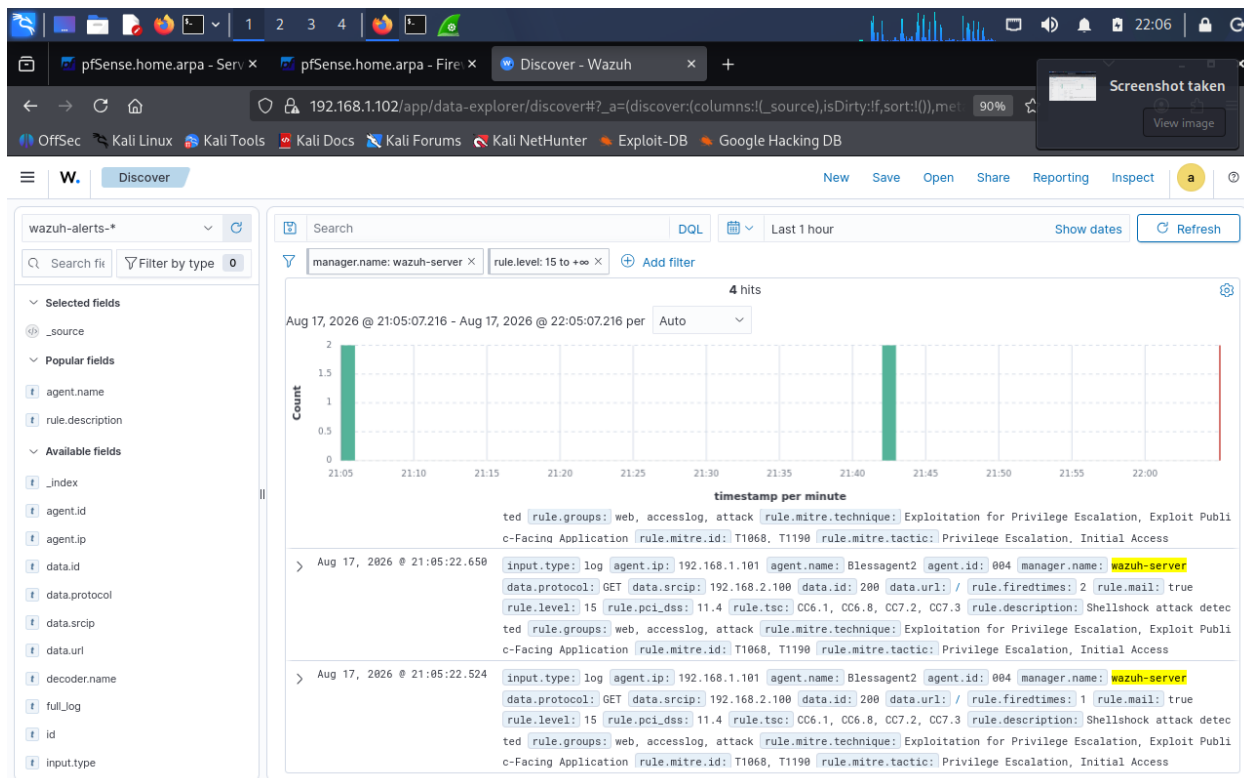
timestamp	agent.name	rule.description	rule.level	rule.id
Aug 17, 2026 @ 21:49:53.6...	Blessagent2	vsftpd: Login failed accessing the FTP server.	5	11403
Aug 17, 2026 @ 21:49:53.6...	Blessagent2	vsftpd: Login failed accessing the FTP server.	5	11403
Aug 17, 2026 @ 21:49:53.6...	Blessagent2	vsftpd: Login failed accessing the FTP server.	5	11403
Aug 17, 2026 @ 21:49:52.0...	Blessagent2	PAM: User login failed.	5	5503
Aug 17, 2026 @ 21:49:51.9...	Blessagent2	PAM: User login failed.	5	5503
Aug 17, 2026 @ 21:49:51.9...	Blessagent2	PAM: User login failed.	5	5503
Aug 17, 2026 @ 21:39:29.8...	Blessagent2	sshd: authentication failed.	5	5760
Aug 17, 2026 @ 21:39:29.4...	Blessagent2	sshd: authentication failed.	5	5760
Aug 17, 2026 @ 21:39:29.4...	Blessagent2	sshd: authentication failed.	5	5760
Aug 17, 2026 @ 21:39:27.4...	Blessagent2	sshd: authentication failed.	5	5760
Aug 17, 2026 @ 21:39:27.4...	Blessagent2	sshd: authentication failed.	5	5760
Aug 17, 2026 @ 21:39:25.4...	Blessagent2	PAM: User login failed.	5	5503
Aug 17, 2026 @ 21:39:25.4...	Blessagent2	PAM: User login failed.	5	5503
Aug 17, 2026 @ 21:39:25.4...	Blessagent2	PAM: User login failed.	5	5503
Aug 17, 2026 @ 21:39:25.4...	Blessagent2	PAM: User login failed.	5	5503

Rows per page: 15

This indicates a multiple authentication alert was detected involving FTP server, PAM user and sshd which means consistent brute force attack.



A review of Wazuh SIEM alerts identified multiple successful SSH authentication and PAM session events on the **Ubuntu server**.



Wazuh detected a **high-severity (Level 15) Shellshock attack** against Ubuntu (**192.168.1.101**). The activity originated from **192.168.2.100** using an HTTP **GET** request. The alert is mapped to **MITRE ATT&CK T1068 (Exploitation for Privilege Escalation)** and **T1190 (Exploit Public-Facing Application)**.

Aug 21 13:51

Wazuh

pfSense.home.arpa - Fire

Not Secure 192.168.1.1/firewall_rules.php?if=lan

VPN

Files

pfSense

COMMUNITY EDITION

System

Interfaces

Firewall

Services

VPN

Status

Diagnostics

Help

WARNING:

The password for this account is insecure. Password is currently set to the default value (pfsense).
Change the password as soon as possible.

Firewall / Rules / LAN

Firewall Rules

Floating

WAN

LAN

DMZ

Rules (Drag to Change Order)

	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☒	0/401 KIB	*	*	*	LAN Address	443 80	*	*		Anti-Lockout Rule	
☐	2/17.68 MiB	IPv4 *	LAN subnets	*	*	*	*	none		Default allow LAN to any rule	
☐	0/0 B	IPv6 *	LAN subnets	*	*	*	*	none		Default allow LAN IPv6 to any rule	

↑ Add

↓ Add

🗑 Delete

🔄 Toggle

📄 Copy

💾 Save

+ Separator

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Rules created on LAN to allow LAN into any rule.

Aug 21 13:50

Wazuh x pfSense.home.arpa - Fire x

Not Secure 192.168.1.1/firewall_rules.php?if=opt1

VPN

System Interfaces Firewall Services VPN Status Diagnostics Help

WARNING:
The password for this account is insecure. Password is currently set to the default value (pfsense).
Change the password as soon as possible.

Firewall / Rules / DMZ

The changes have been applied successfully. The firewall rules are now reloading in the background.
Monitor the filter reload progress.

Floating WAN LAN **DMZ**

Rules (Drag to Change Order)

	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	0/448 B	IPv4 TCP	192.168.2.100	*	192.168.1.101	22 (SSH)	*	none		Blocked ssh traffic	
☐	0/59 KiB	IPv4 *	DMZ subnets	*	*	*	*	none		Default block DMZ to any rule	
☐	0/0 B	IPv4 TCP	192.168.2.100	*	192.168.1.101	*	*	none		blocked malicious IP to internet access to LAN	

Add Add Delete Toggle Copy Save Separator

Rules on DMZ to block malicious IP (kali).

```
Session [x] Terminal Emulator Use the command line kali@kali: ~
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group def link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:5a:87:bc brd ff:ff:ff:ff:ff:ff
    inet 192.168.2.100/24 brd 192.168.2.255 scope global dynamic noprefixroute
        valid_lft 6860sec preferred_lft 6860sec
    inet6 fe80::ee9b:8888:db3e:c03/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

(kali@kali)~$ ping 192.168.1.101
PING 192.168.1.101 (192.168.1.101) 56(84) bytes of data:
^C
    192.168.1.101 ping statistics:
    18 packets transmitted, 0 received, 100% packet loss, time 19792ms
    or too busy. Try again in a few moments.

(kali@kali)~$
```

Kali can not reach ubuntu any longer after the rules have been applied.

```
Aug 21 13:27
ubuntu@ubuntu-VirtualBox: ~
ubuntu@ubuntu-VirtualBox:~$ ping 192.168.2.100
PING 192.168.2.100 (192.168.2.100) 56(84) bytes of data.
64 bytes from 192.168.2.100: icmp_seq=1 ttl=63 time=0.851 ms
64 bytes from 192.168.2.100: icmp_seq=2 ttl=63 time=0.718 ms
64 bytes from 192.168.2.100: icmp_seq=3 ttl=63 time=0.846 ms
64 bytes from 192.168.2.100: icmp_seq=4 ttl=63 time=0.677 ms
64 bytes from 192.168.2.100: icmp_seq=5 ttl=63 time=0.893 ms
64 bytes from 192.168.2.100: icmp_seq=6 ttl=63 time=0.710 ms
64 bytes from 192.168.2.100: icmp_seq=7 ttl=63 time=1.00 ms
64 bytes from 192.168.2.100: icmp_seq=8 ttl=63 time=0.926 ms
64 bytes from 192.168.2.100: icmp_seq=9 ttl=63 time=0.950 ms
64 bytes from 192.168.2.100: icmp_seq=10 ttl=63 time=1.01 ms
64 bytes from 192.168.2.100: icmp_seq=11 ttl=63 time=1006 ms
64 bytes from 192.168.2.100: icmp_seq=12 ttl=63 time=0.766 ms
^C
--- 192.168.2.100 ping statistics ---
12 packets transmitted, 12 received, 0% packet loss, time 11121ms
rtt min/avg/max/mdev = 0.677/84.575/1005.556/277.686 ms
ubuntu@ubuntu-VirtualBox:~$
```

While Ubuntu(LAN) can reach DMZ(kali) after the rule.

Aug 21 13:35

Wazuh x pfSense.home.arpa - Stat x

Not Secure 192.168.1.1/status_logs_filter.php

VPN

✗	Aug 21 11:00:01	DMZ	Default block DMZ to any rule (1785937190)	i 192.168.2.100:42054	i 192.168.1.1:443	TCP:S
✗	Aug 21 11:00:00	DMZ	Default block DMZ to any rule (1785937190)	i 192.168.2.100:50940	i 192.168.2.1:53	UDP
✗	Aug 21 11:00:00	DMZ	Default block DMZ to any rule (1785937190)	i 192.168.2.100:50940	i 192.168.2.1:53	UDP
✗	Aug 21 11:00:00	DMZ	Default block DMZ to any rule (1785937190)	i 192.168.2.100:53792	i 192.168.2.1:53	UDP
✗	Aug 21 11:00:00	DMZ	Default block DMZ to any rule (1785937190)	i 192.168.2.100:57779	i 192.168.2.1:53	UDP
✗	Aug 21 11:00:00	DMZ	Default block DMZ to any rule (1785937190)	i 192.168.2.100:37415	i 192.168.2.1:53	UDP
✗	Aug 21 11:00:00	DMZ	Default block DMZ to any rule (1785937190)	i 192.168.2.100:37415	i 192.168.2.1:53	UDP
✗	Aug 21 11:00:00	DMZ	Default block DMZ to any rule (1785937190)	i 192.168.2.100:51704	i 192.168.2.1:53	UDP
✗	Aug 21 11:00:00	DMZ	Default block DMZ to any rule (1785937190)	i 192.168.2.100:51704	i 192.168.2.1:53	UDP
✗	Aug 21 11:00:00	DMZ	Default block DMZ to any rule (1785937190)	i 192.168.2.100:45281	i 192.168.2.1:53	UDP
✗	Aug 21 10:59:59	DMZ	Default block DMZ to any rule (1785937190)	i 192.168.2.100:32960	i 192.168.2.1:53	UDP
✗	Aug 21 10:59:59	DMZ	Default block DMZ to any rule (1785937190)	i 192.168.2.100:32960	i 192.168.2.1:53	UDP
✗	Aug 21 10:59:59	DMZ	Default block DMZ to any rule (1785937190)	i 192.168.2.100:33631	i 192.168.2.1:53	UDP
✗	Aug 21 10:59:59	DMZ	Default block DMZ to any rule (1785937190)	i 192.168.2.100:33631	i 192.168.2.1:53	UDP
✗	Aug 21 10:59:59	DMZ	Default block DMZ to any rule (1785937190)	i 192.168.2.100:43737	i 192.168.2.1:53	UDP
✗	Aug 21 10:59:59	DMZ	Default block DMZ to any rule (1785937190)	i 192.168.2.100:43737	i 192.168.2.1:53	UDP
✗	Aug 21 10:59:59	DMZ	Default block DMZ to any rule (1785937190)	i 192.168.2.100:37368	i 192.168.2.1:53	UDP
✗	Aug 21 10:59:59	DMZ	Default block DMZ to any rule (1785937190)	i 192.168.2.100:37368	i 192.168.2.1:53	UDP
✗	Aug 21 10:59:58	DMZ	Default block DMZ to any rule (1785937190)	i 192.168.2.100:44447	i 192.168.2.1:53	UDP
✗	Aug 21 10:59:58	DMZ	Default block DMZ to any rule (1785937190)	i 192.168.2.100:44447	i 192.168.2.1:53	UDP

Firewall shows that DMZ cannot reach LAN any longer because of the rules passed.